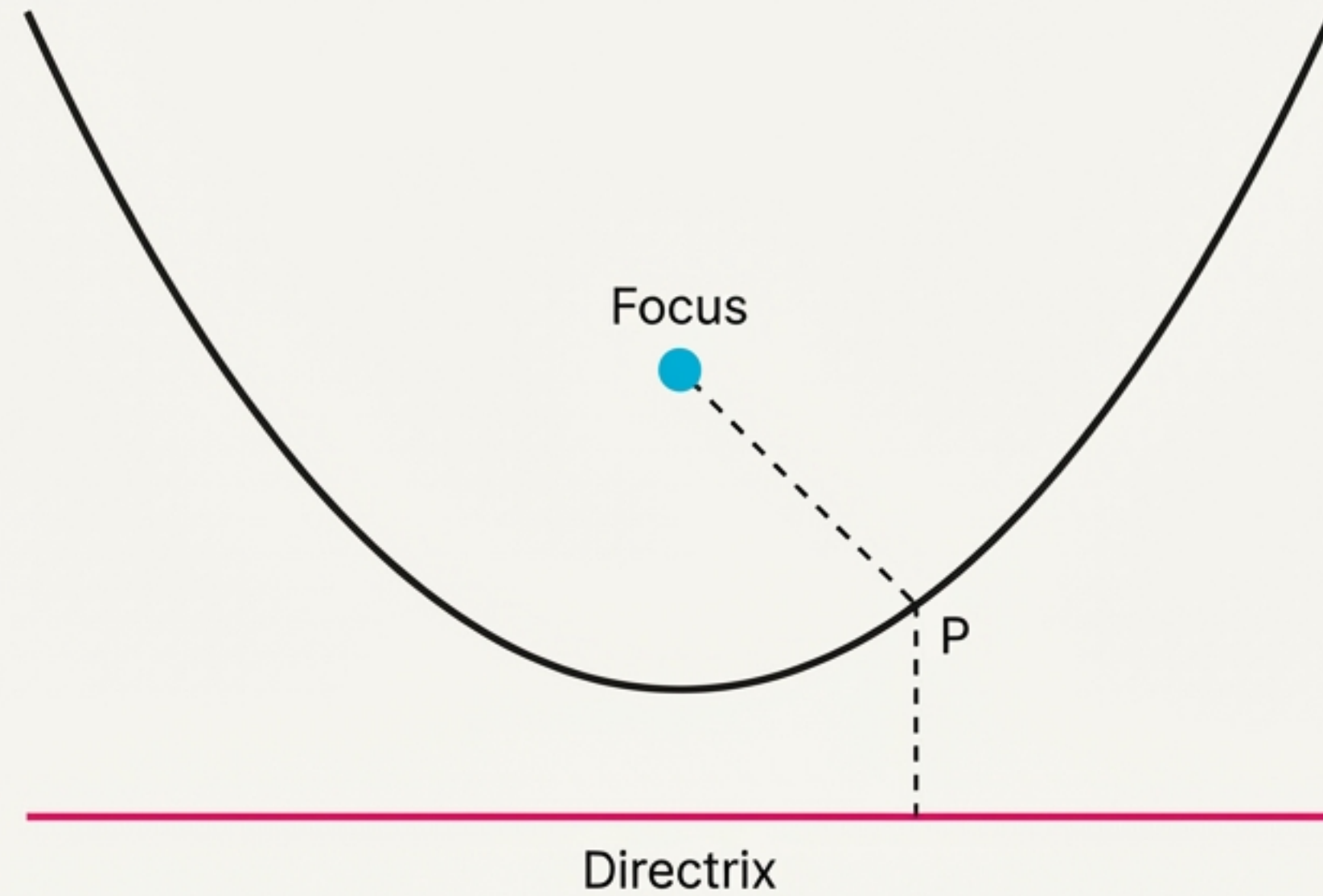




The Geometry of the Parabola

Deriving the algebraic shape from the Focus and Directrix.

We often encounter parabolas as equations ($y=x^2$). But before they were algebra, they were geometry. This deck derives the mathematical proof that a set of points equidistant from a single point and a line creates the perfect U-shape we know as a parabola.

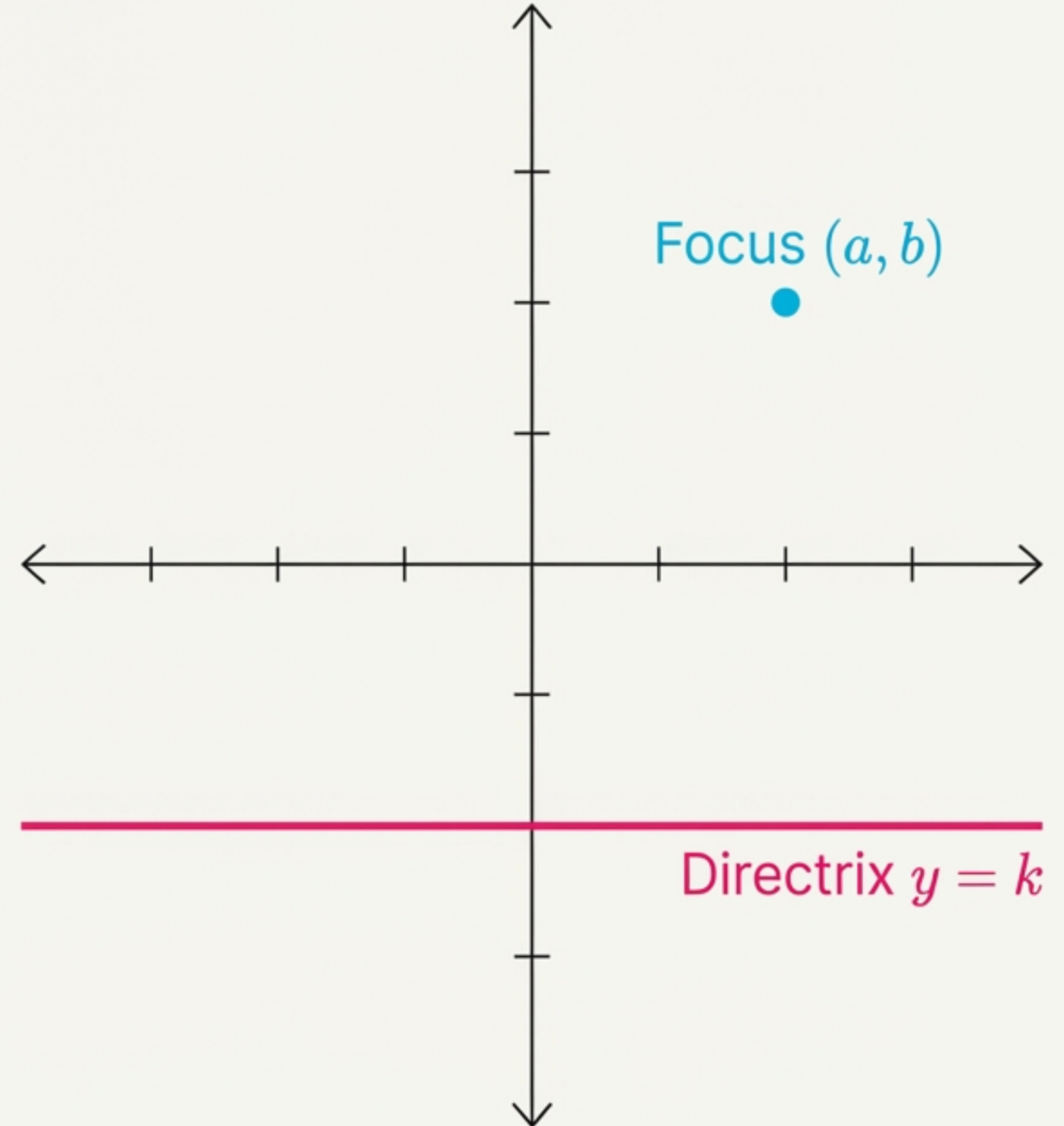
The Definitions

To build the proof, we must define the players.

The **Focus (Cyan)**: A fixed point in the plane with coordinates (a, b) .

The **Directrix (Magenta)**: A fixed horizontal line with the equation $y = k$.

The Goal: We want to find the 'Locus'—the specific set of all points (x, y) that are exactly the same distance from the Focus as they are from the Directrix.

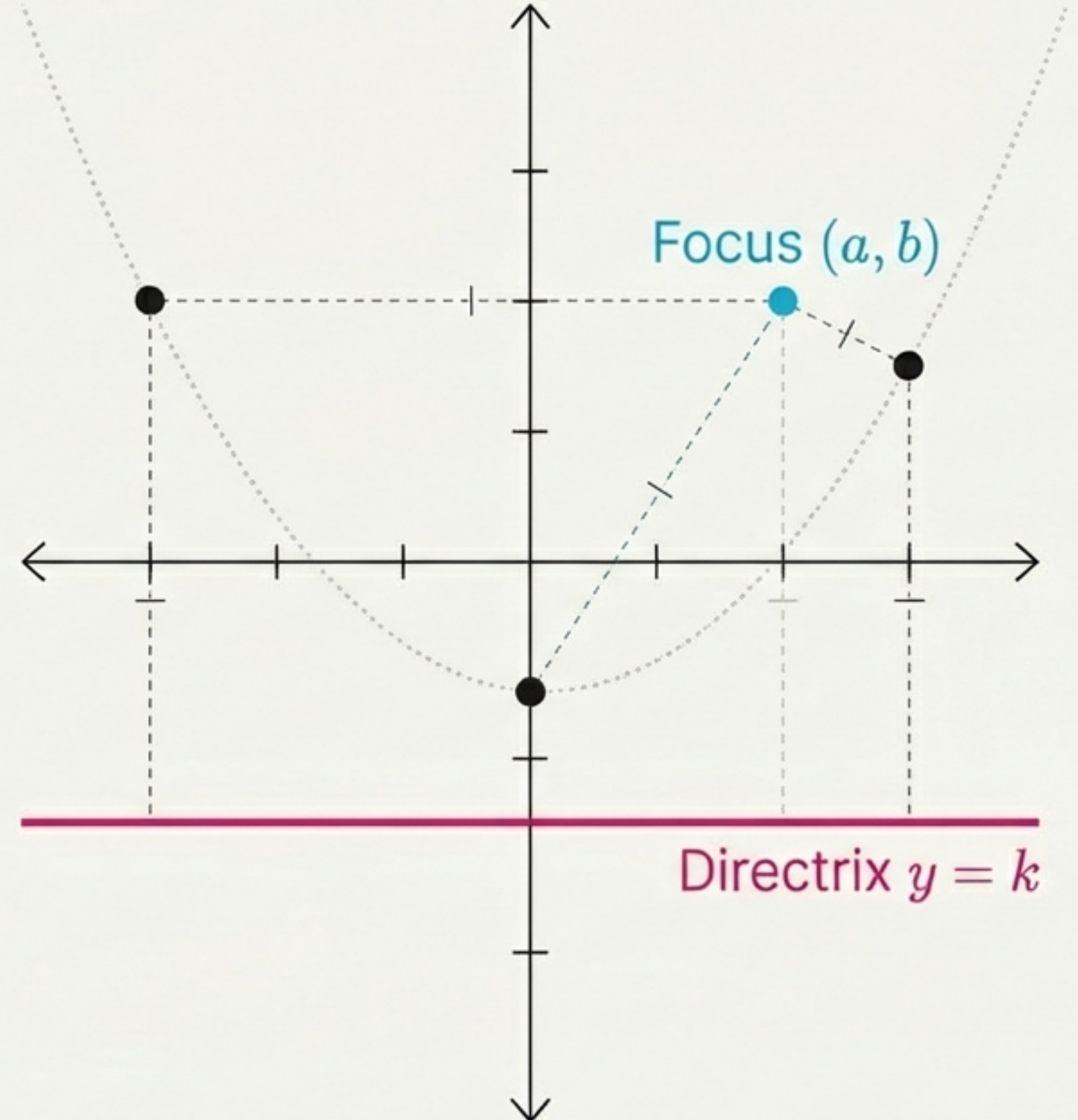


Visualizing the Locus

If we plot points that satisfy the rule 'Distance to Focus = Distance to Directrix,' a pattern emerges.

1. The vertex is clearly halfway between the point and the line.
2. As we move sideways, we must move up to maintain equality.

The Intuition: Connecting these points reveals a shape that looks suspiciously like a parabola. Now, we must prove it.



The Mathematical Challenge

To prove this shape is a parabola, we must express the geometric relationship algebraically.

Let $P(x, y)$ be any point in the locus.

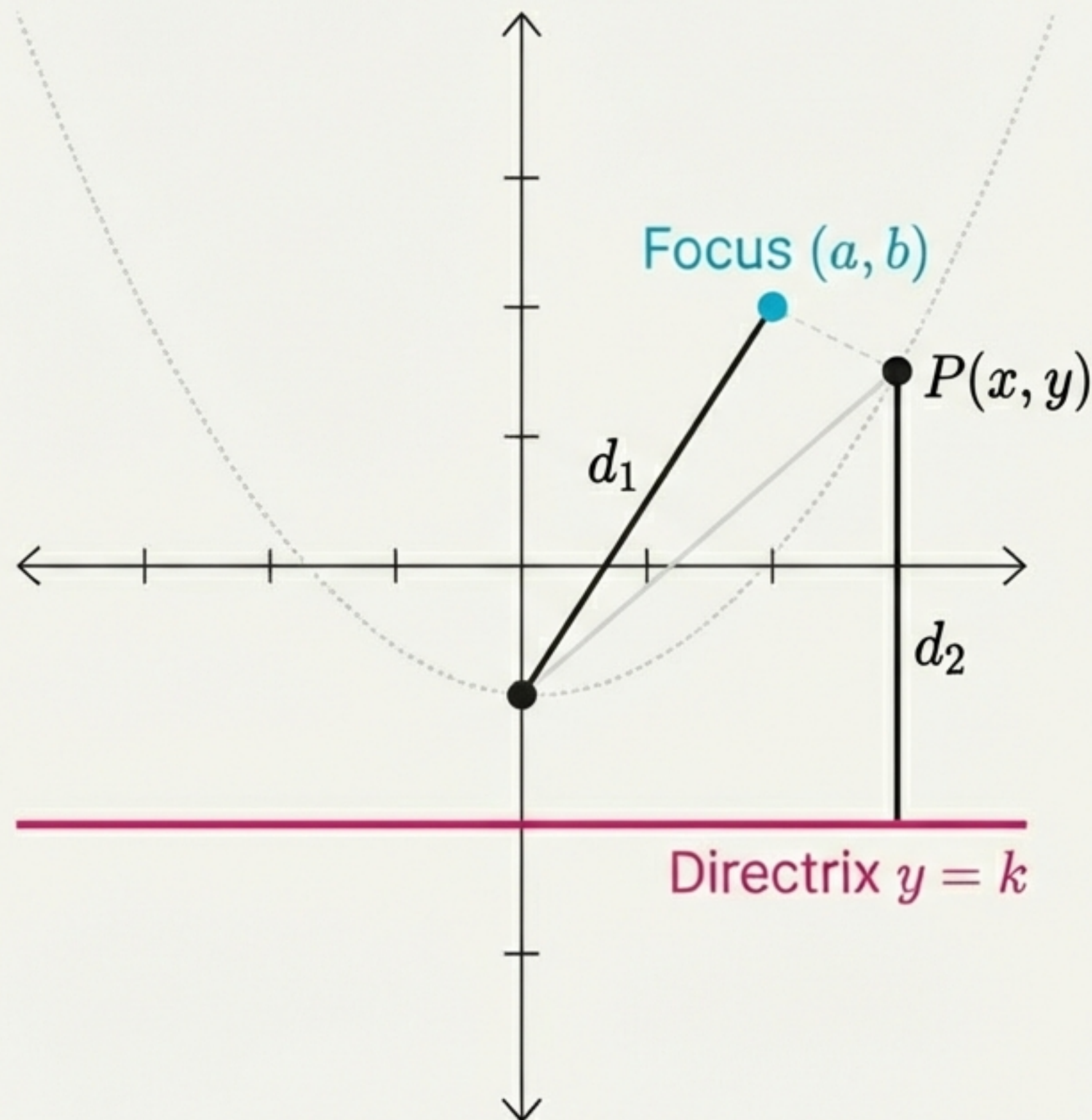
Let d_1 = Distance from P to the Focus.

Let d_2 = Distance from P to the Directrix.

The Law: $d_1 = d_2$

Tracker:

Step 1: Define Distances



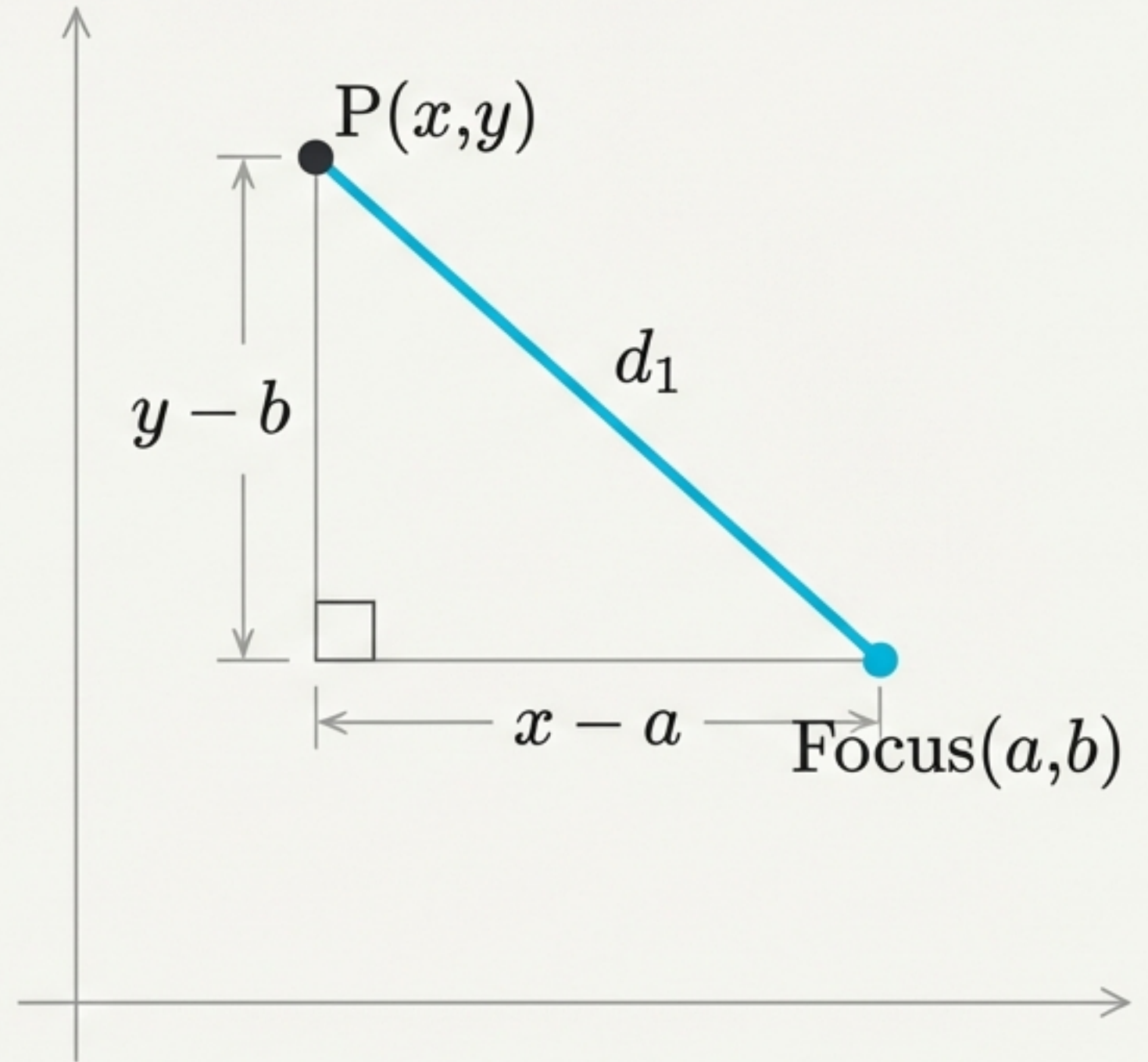
Distance to the Focus (d_1)

Using the Pythagorean distance formula between point (x, y) and Focus (a, b) :

$$d_1 = \sqrt{(x - a)^2 + (y - b)^2}$$

Note: We square the difference in x coordinates and the difference in y coordinates.

Tracker: Step 1: Define Distances

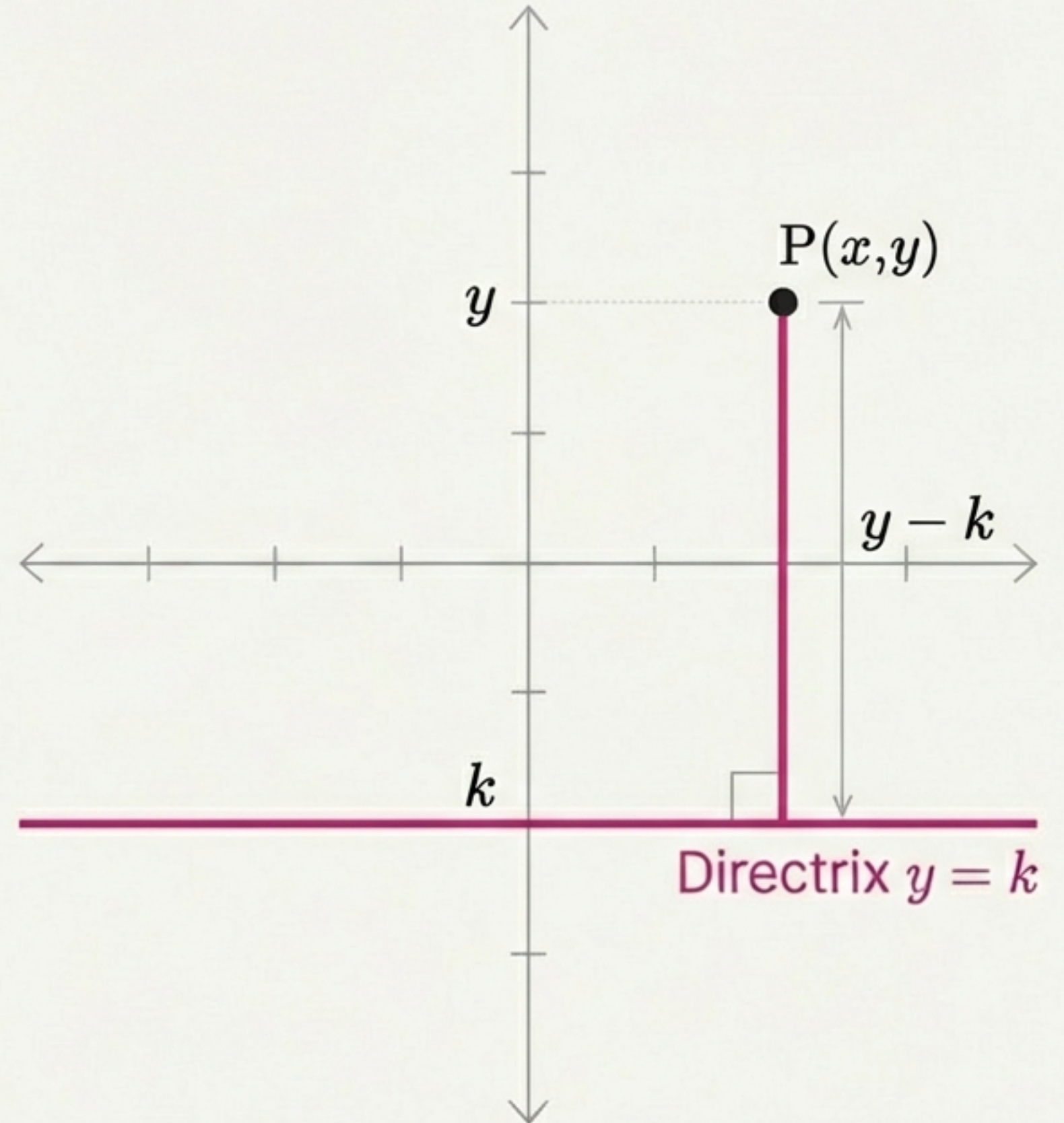


Distance to the Directrix (d_2)

The shortest distance from a point to a horizontal line is a straight vertical drop. The difference is simply $(y - k)$. To ensure the result is positive and to match the format of d_1 , we write it as:

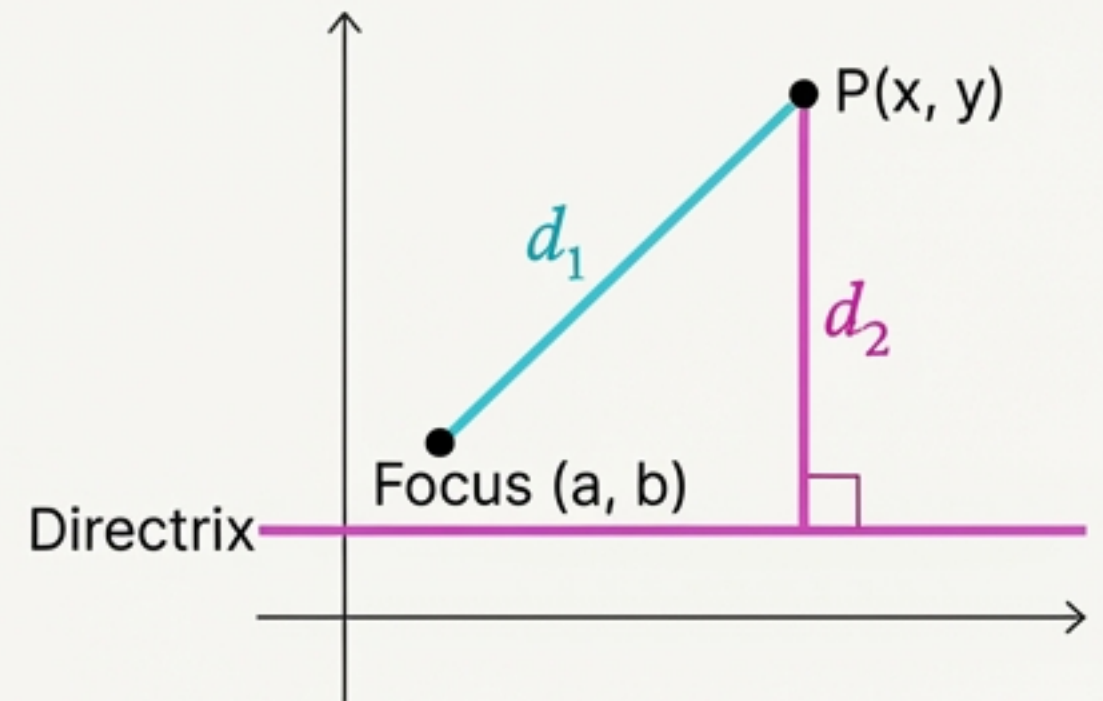
$$d_2 = \sqrt{(y - k)^2}$$

Tracker: Step 1: Define Distances



The Equality

We set $d_1 = d_2$ to enforce the geometric rule.
This is our Master Equation.



$$\sqrt{(x - a)^2 + (y - b)^2} = \sqrt{(y - k)^2}$$

Observation: This contains x^2 and y^2 terms inside radicals. It does not yet look like a parabola.

Tracker:

Step 1: Define Distances
[Complete]

Eliminating the Radicals

Radicals make algebra difficult. Squaring both sides removes the square roots entirely.

Tracker: Step 2:
Square Both Sides

$$\sqrt{(x - a)^2 + (y - b)^2} = \sqrt{(y - k)^2}$$



Square Both Sides

$$(x - a)^2 + (y - b)^2 = (y - k)^2$$

The Expansion

We expand the $(y-b)^2$ and $(y-k)^2$ terms. We leave the $(x-a)^2$ term alone, as we want to keep the x-variable compact.

Tracker: Step 3: Expansion

$$(x - a)^2 + (y - b)^2 = (y - k)^2$$

$$(x - a)^2 + (y^2 - 2yb + b^2) = (y^2 - 2yk + k^2)$$

The Cleanup

Notice that we have a positive y^2 on both the left and right sides. Subtracting y^2 from both sides makes them vanish.

Significance

The y^2 is gone. We are now left with an equation where x is squared, but y is linear. This is the definition of a parabola.

$$(x - a)^2 + \cancel{y^2} - 2yb + b^2 = \cancel{y^2} - 2yk + k^2$$

$$(x - a)^2 - 2yb + b^2 = -2yk + k^2$$

Tracker:

Step 4: Cancellation

Grouping the Terms

We want to isolate y to match standard form.

1. Add $2yb$ to both sides.
2. Subtract b^2 from both sides.

This groups all y terms on the right side.

$$(x - a)^2 = 2yb + 2yk + b^2 + k^2$$
$$(x - a)^2 = 2yb - 2yk - b^2 + k^2 + b^2$$

y terms constants

Factoring

We factor out $2y$ from the terms. We also recognize $(b^2 - k^2)$ as a difference of squares $(b - k)(b + k)$, though here we are subtracting them, so we manipulate signs carefully to prepare for division.

$$(x - a)^2 = y(2b - 2k) - (b^2 - k^2)$$

$$(x - a)^2 + (b^2 - k^2) = y(2b - 2k)$$

$$(x - a)^2 + (b - k)(b + k) = 2y(b - k)$$

The Derived Equation

Divide the entire equation by $2(b-k)$ to isolate y . We have successfully converted the geometric rule into a functional equation.

$$y = \frac{1}{2(b-k)}(x-a)^2 + \frac{b+k}{2}$$

Tracker:

Step 6: Final Form [Complete]

Pattern Matching

This complex result maps perfectly to the standard Vertex Form:

The mess of constants is simply A (Width).
The Focus coordinate 'a' is simply h (Horizontal shift).

The constant term is simply v (Vertical shift).

Conclusion: The locus of points equidistant from a Focus and Directrix is, undeniably, a parabola.

$$y = \frac{1}{2(b-k)}(x-a)^2 + \frac{b+k}{2}$$

=

$$y = A(x-h)^2 + v$$

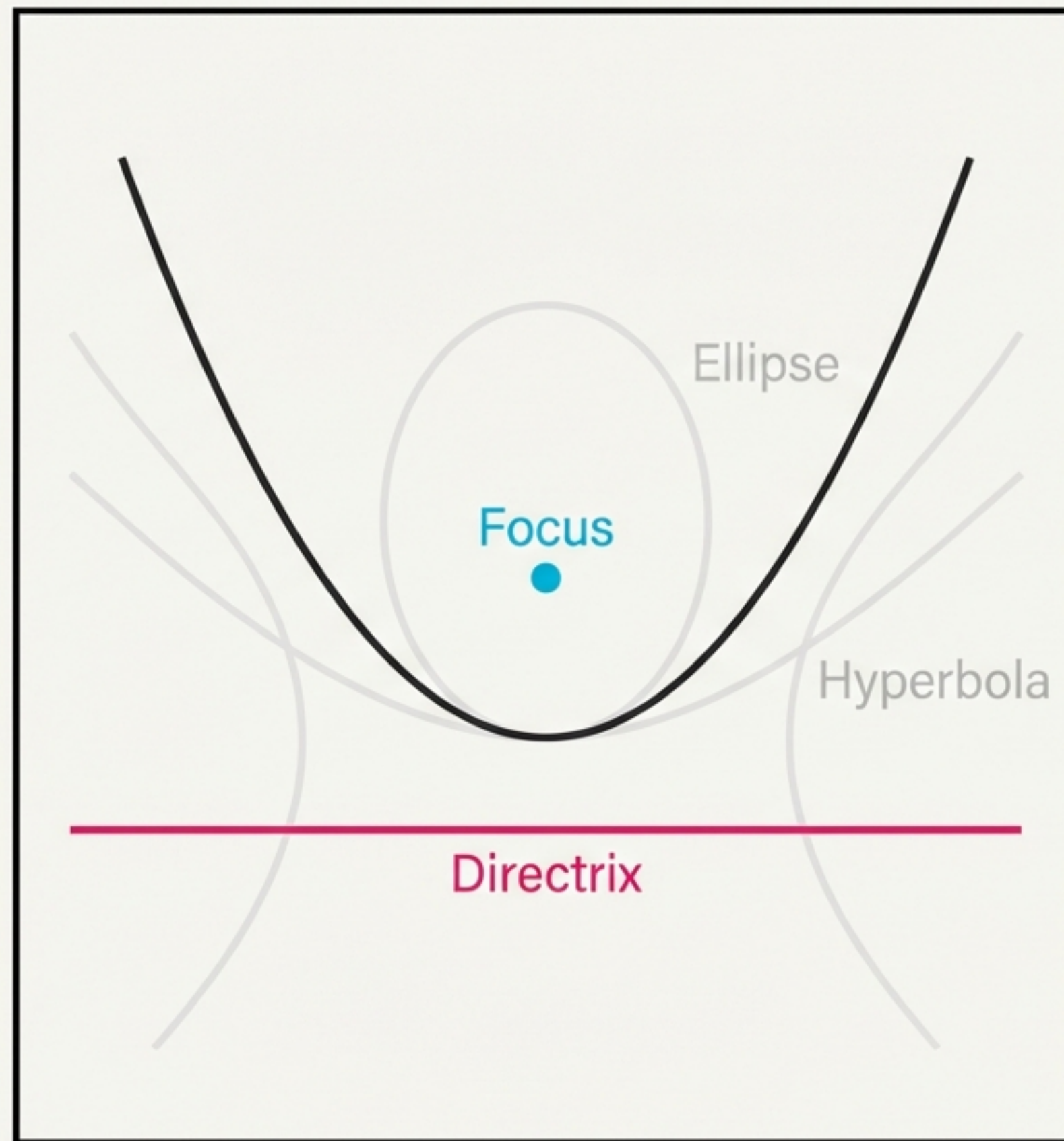
Beyond the Parabola

We have proven that a parabola is the perfect balance (ratio 1:1) between a point and a line. But what if that ratio changes?

If points are closer to the focus (ratio < 1), we get an Ellipse.

If points are further from the focus (ratio > 1), we get a Hyperbola.

This derivation is the gateway to understanding all Conic Sections.



Tracker: Step 8: Beyond the Parabola [Expansion]